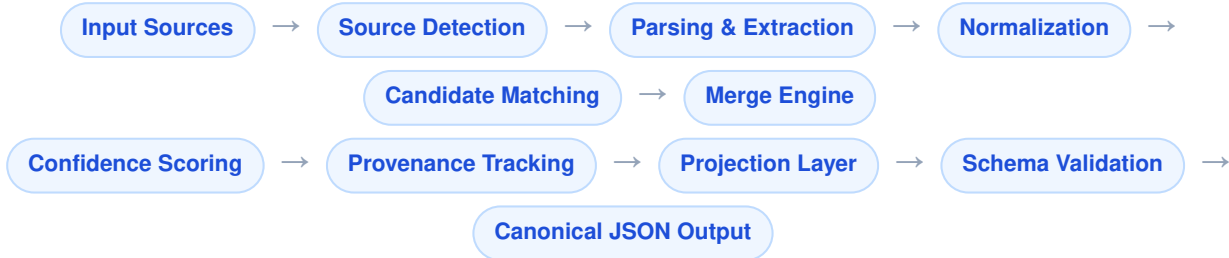


Multi-Source Candidate Data Transformer

Eightfold Engineering Internship Assignment (Jul–Dec 2026)

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1. PIPELINE ARCHITECTURE



2. CANONICAL SCHEMA

- candidate_id
- full_name
- emails
- phones (E.164)
- location (ISO-3166)
- links
- headline
- years_experience
- skills
- experience
- education
- provenance
- overall_confidence

4. MERGE POLICY

Candidate Matching Keys:

- Candidate ID
- Email
- Phone

Conflict Resolution Priority:

Structured Sources → Resume → Recruiter Notes

Note: Duplicate values are aggressively removed.

3. NORMALIZATION STRATEGY

- Phones** → E.164 using phonenumbers
- Dates** → YYYY-MM using dateparser
- Location** → ISO-3166 country codes
- Skills** → Canonical mapping & synonym resolution

5. CONFIDENCE STRATEGY

- High** Same value found in multiple trusted sources
- Med** Single trusted source
- Low** Conflicting values across sources

6. CONFIGURABLE OUTPUT

Support:

- ✓ Field Selection
- ✓ Field Renaming
- ✓ Toggle Provenance
- ✓ Toggle Confidence

Missing Value Strategy:

- null: Emit key with null value
- omit: Drop key entirely
- error: Raise validation exception

7. EDGE CASES HANDLED

- Missing source files
- Corrupted PDF payloads
- Duplicate candidate entries
- Multiple phone formats
- Varying date semantics
- Missing optional fields

8. TECHNOLOGIES

Python Pandas Pydantic PyMuPDF Dateparser
Phonenumbers